



# Performance analysis of smartphone gnss mobile applications for fit-for-purpose cadastral surveying

Oluibukun Gbenga Ajayi<sup>1,2,3</sup> · Gideon Abidemi Aleji<sup>3</sup> · Oluwatobi Olalekan Adetunji<sup>4</sup> · Yusuf Drisu Opaluwa<sup>3</sup>

Received: 14 April 2025 / Accepted: 1 September 2026  
© The Author(s) 2026

## Abstract

The growing need for affordable and accessible geospatial tools in land administration has intensified interest in mobile GNSS applications, particularly for Fit-For-Purpose (FFP) Cadastral Mapping. This study evaluates the positional accuracy of eight selected mobile GNSS applications by comparing their coordinate outputs against reference data obtained from Differential GNSS (DGNSS). Initially, ten applications were tested under similar environmental conditions; however, two were excluded from final analysis due to persistent extreme deviations and irregular positional stability across the survey stations. Their Root Mean Square Error (RMSE) values were significantly higher than those of the other applications and exceeded the acceptable accuracy threshold for FFP Cadastral Mapping. The remaining eight apps were assessed using statistical methods, including the Kruskal-Wallis test, to determine significant differences in positional accuracy. *The Mobile Topography application* demonstrated the highest accuracy, with a mean deviation of 2.83 m, a 95% confidence interval of 1.57–4.09, and a standard deviation of 1.20 m. Aside from the descriptive statistics, the RMSE was computed to provide an overall measure of positional accuracy relative to the DGPS reference coordinates. *Mobile Topography* had the lowest RMSE of 3.03 m, falling within the acceptable FFP land administration range of not more than 5 m in rural areas. *Polaris* and *GPS Essentials* applications showed moderate reliability, while other applications recorded deviations well beyond acceptable limits, making them unsuitable for boundary-sensitive cadastral tasks. The findings affirm the potential of mobile GNSS apps as cost-effective tools for FFP mapping in resource-limited contexts while also highlighting their variability and the necessity of integrating error-mitigation strategies such as post-processing and field validation. Future research will focus on improving accuracy through hardware optimization, algorithmic corrections, and robust field methodologies to enhance the viability of mobile GNSS in FFP cadastral systems.

**Keywords** Fit-for-purpose (FFP) cadastral mapping · Mobile GNSS applications · Differential GNSS (DGNSS) · Positional accuracy · Land administration · Affordable mapping solutions

## Introduction

Effective land management is central to livelihood opportunities, land rights, juridical security, and the orderly development and administration of cities and settlements. An efficient, adaptable, and socially responsive land administration system supports cadastral procedures and also contributes to tenure security, reduced conflict, and more equitable access to land-related services (Mutale and Ajayi 2026). The impact of such systems on national stability is evident in countries like Colombia, where efficient land administration has contributed to lasting peace (Chigbu et al. 2021). Given their critical role in social and economic development, land management institutions must adopt flexible and efficient approaches that can support practical land administration

✉ Oluibukun Gbenga Ajayi  
oluibukun.ajayi@up.ac.za

<sup>1</sup> Department of Geography, Geoinformatics and Meteorology, University of Pretoria, Pretoria, South Africa

<sup>2</sup> Department of Land and Spatial Sciences, Namibia University of Science and Technology, Windhoek, Namibia

<sup>3</sup> Department of Surveying and Geoinformatics, Federal University of Technology, Minna, Nigeria

<sup>4</sup> Department of Geodetic Science, African Institute for Geospatial Information Science and Technology, Ile-Ife, Nigeria

solutions at scale. Apostolopoulos et al. (2018) highlighted ongoing global efforts to develop tools and methodologies to reduce costs and simplify cadastral procedures. They further emphasized the potential for increased involvement of property owners and local citizens in cadastral mapping, an approach referred to as crowdsourcing in land administration, which allows tasks previously reserved for professional Surveyors and government agencies to be performed by the public.

Traditional land administration systems in developing countries have often been inefficient due to high implementation costs, complex procedures, limited coverage, the exclusion of informal landholders, high technical demands, and slow adaptation to evolving technologies. These weaknesses affect cadastral delivery and the broader goals of secure land rights, effective land-use control, and inclusive land governance. To address these inefficiencies, the World Bank and the International Federation of Surveyors (FIG) recommended the Fit for Purpose Land Administration (FFPLA) model (Bennett and Unger 2021). FFPLA is designed to serve the fundamental purposes of land management, securing tenure for all and controlling land use, without relying on high-cost and technology-intensive procedures. Instead, the approach prioritizes affordability, accessibility, and incremental improvements in accuracy over time, based on societal and user needs. FFPLA consists of three key frameworks: spatial, legal, and institutional. The spatial framework integrates aerial or satellite imagery for large-scale cadastral mapping, participatory field adjudication, and progressive accuracy enhancement. Many developing countries have begun adopting these methods, particularly aerial and satellite imagery, though cost constraints remain a significant challenge. To further promote public participation in data collection, countries like Mozambique and Colombia have turned to more affordable field data acquisition methods, such as GNSS-integrated mobile applications (Chigbu et al. 2021).

Differential Global Navigation Satellite Systems (DGNSS) are widely regarded as the most efficient tool for land administration due to their high level of accuracy in field data acquisition (Taiwo et al. 2025). However, this preference can sit uneasily with the cost-effective and participatory principles of FFPLA, since DGNSS requires expensive equipment and specialised technical expertise. DGNSS, which often relies on signals from multiple GNSS constellations, offers sub-centimetre positioning accuracy, making it superior to traditional optical instruments like theodolites and total stations. Despite its advantages, the high cost and technical complexity of DGNSS limit its accessibility to the public, thereby reducing community participation in data collection, which is an essential component of FFPLA (Bennett and Unger 2021).

The FFPLA approach recognises that spatial accuracy should be purpose-driven rather than dictated by rigid technical standards. Research suggests that an accuracy range of 2 to 5 m is generally acceptable for rural and general FFPLA applications (Taiwo et al. 2025; Rocha et al. 2021). This level of precision is sufficient for many parcel mapping tasks while keeping costs low and participation broad. Recent advances in technology have made position fixing more accessible through smartphone GNSS capabilities, which are increasingly embedded in ordinary mobile devices. This creates an important opportunity for strengthening the spatial framework of FFPLA, especially because many landowners and professionals already own smartphones and can use GNSS-enabled applications with relative ease. Despite the growing emphasis on public participation in land administration, limited research has explored the suitability of smartphone positioning as a practical, cost-effective, and user-friendly tool for cadastral mapping. This study, therefore, assesses the accuracy and applicability of smartphone GNSS positioning for FFPLA and identifies the most suitable mobile application for this purpose.

## Review of relevant literature

Fit-for-Purpose Land Administration (FFPLA) has emerged as a practical and cost-effective approach to land administration, particularly in developing countries where conventional surveying methods are often constrained by high costs, lengthy implementation timelines, and limited accessibility. By embracing the principle of “good enough” spatial accuracy, flexible legal frameworks, and incremental system improvements, FFPLA enables approaches such as participatory mapping, aerial imagery-based boundary delineation, and simplified adjudication procedures that significantly reduce implementation costs and delays while maintaining key legal principles such as Torrens indefeasibility (Asman et al. 2025). The FFP approach prioritizes practical solutions that are sufficient for land tenure security, rather than strict adherence to high-precision standards that may be unnecessary for many land administration purposes.

Recent studies have emphasized the viability of alternative cadastral mapping techniques, particularly aerial imagery and remotely sensed data, as effective substitutes for conventional ground-based methods. Compared high-resolution aerial imagery with conventional surveying techniques such as total station, theodolite traverse, and Real-Time Kinematic (RTK) GNSS in Ekiti State, Nigeria. Their findings indicate that aerial imagery-based cadastral mapping provides comparable accuracy while offering broader coverage and faster data acquisition. This suggests that FFP approaches can significantly expedite land

registration processes without compromising spatial accuracy. Similarly, Taiwo et al. (2024) investigated the suitability of various remotely sensed data resolutions for cadastral mapping across rural, peri-urban, and urban settlements in Ekiti State, Nigeria. Their study found that high-resolution aerial imagery (0.05 m to 1 m) is particularly effective at delineating property boundaries, enabling land tenure security at a cost-effective level.

Also, UAV technology, in particular, has been highlighted as a flexible and scalable tool that aligns well with FFP principles. Ajayi and Oruma (2022a, b) developed a hybrid approach that couples UAV photogrammetry with automatic feature extraction to modernize cadastral mapping processes. The article detailed how UAV-derived high-resolution imagery enhances the detection of parcel boundaries and other land features, thereby reducing the cost and time compared to traditional surveying methods. The study demonstrated that integrating these advanced techniques can significantly increase mapping accuracy and reliability. Emphasis is placed on overcoming challenges in the automatic extraction of cadastral features, making a strong case for the adoption of technology-driven solutions in enhancing land administration systems. Further, Ajayi et al. (2021) and Ajayi et al. (2018) developed a dedicated geospatial information system tailored to processing and managing cadastral survey data. The proposed software integrates data capture, processing, and storage functionalities specifically for cadastral applications. It is designed to streamline workflows from field data collection to digital land record management, thereby addressing limitations in conventional cadastral systems. The authors argue for a practical, scalable digital land administration system that accommodates varying accuracy and completeness levels based on local needs. The developed tools emphasized ease of integration with existing cadastral frameworks while facilitating rapid updating of land records.

Beyond technical feasibility, quality assurance remains a crucial factor in the adoption of FFP land administration systems. Rocha et al. (2021) emphasized the importance of implementing quality control mechanisms to ensure that FFP mapping solutions meet minimum accuracy standards. Their study proposed a structured quality assurance model that enhances the reliability of spatial data collected using FFP methodologies, reinforcing their credibility for legal and administrative purposes.

These studies collectively highlight the increasing recognition of FFP cadastral mapping as a practical alternative to traditional surveying techniques, especially in resource-limited settings. The integration of UAVs, aerial imagery, and quality assurance frameworks has significantly advanced land tenure documentation while reducing financial and technical constraints. However, despite the benefits of aerial

mapping, it does not fully align with the cost-effectiveness and efficiency objectives of FFP cadastral mapping. The core principle of FFP mapping is to actively involve landholders and the general public in crowdsourcing land information through affordable and accessible methods. Aerial surveys, though effective, remain expensive, prompting researchers to explore the feasibility of smartphone positioning as a more economical and scalable solution for FFP cadastral mapping.

### Smart phone positioning apps for FFP cadastral mapping

The rapid advancement of smartphone technology has opened new possibilities for cost-effective land administration solutions, particularly in Fit-for-Purpose (FFP) cadastral mapping. Modern smartphones, equipped with Global Navigation Satellite System (GNSS) receiver capabilities, have become potential tools for land tenure documentation in resource-constrained settings. Several recent studies have examined the feasibility, accuracy, and limitations of smartphone positioning in cadastral surveying.

Smartphone GNSS receivers have improved significantly with the introduction of dual-frequency, multi-constellation capabilities. A study by Retscher and Weigert (2022) assessed the Google Pixel 5's GNSS receiver and found that, while it could achieve sub-meter accuracy with extended observation periods, its data quality limited its usability for high-precision applications. This study suggested that smartphones can be suitable for general GIS data collection but may not yet replace survey-grade GNSS for precise cadastral work. Further exploring the feasibility of smartphone GNSS, Hernández et al. (2024) proposed a cloud-based real-time processing approach to enhance the positioning accuracy of mobile GNSS observations. Their study demonstrated that offloading computational tasks to cloud platforms improves navigation accuracy, making smartphones more viable for various surveying applications, including cadastral mapping.

Multiple studies have compared smartphone GNSS accuracy to that of professional-grade GNSS receivers. Streach et al. (2024) analyzed the effectiveness of Real-Time Kinematic (RTK)-enabled smartphones in areas where RTK signals were obstructed. They found that while smartphones collected a large number of observations, their accuracy in RTK-denied environments was significantly lower than traditional survey methods, reinforcing the need for correction techniques. Similarly, a comparative study by Oluwadare and Salami (2024) evaluated parcel boundary determinations using smartphones and survey-grade GNSS receivers. Their findings indicated that while smartphones provided a cost-effective alternative, their accuracy remained inferior

to differential GNSS (DGNSS) methods, making them less suitable for applications requiring high precision.

Environmental conditions significantly affect smartphone GNSS accuracy, particularly in urban and forested areas. Huang et al. (2022) examined the positioning performance of smartphones under varying canopy openness. They concluded that dual-frequency smartphones, when corrected with Precise Point Positioning (PPP) techniques, achieved horizontal accuracies of approximately 2.5 m in open environments but struggled in dense canopy conditions, where survey-grade GNSS remained necessary.

To address urban positioning challenges, Amir and Mohammad (2024) integrated extended Kalman filters and inertial navigation systems with smartphone GNSS to mitigate multipath errors. Their study showed that while smartphones were affected by signal obstructions from buildings, accuracy could be improved through algorithmic corrections, increasing their viability for cadastral mapping in urban settings.

These studies collectively suggest that while smartphone GNSS technology presents a promising and cost-effective alternative for FFP cadastral mapping, its accuracy remains variable. Factors such as environmental conditions, hardware limitations, and data correction techniques influence their reliability. The integration of real-time cloud computing, advanced filtering algorithms, and hybrid positioning methods could further enhance smartphone accuracy, aligning with FFP principles of affordability and inclusivity. However, no sufficient studies have been conducted to validate the applicability of GNSS mobile applications specifically for FFP cadastral mapping, where high-end accuracy is not required. Furthermore, there has been no comprehensive study comparing smartphone GNSS positioning results against FFP cadastral accuracy standards, which would lend scientific credibility and support the objectives of FFP cadastral mapping. Additionally, no research has systematically compared the available GNSS mobile applications to determine which is most suitable for FFP cadastral mapping.

### Accuracy standards for FFP cadastral mapping

Establishing appropriate accuracy standards is crucial for implementing Fit-For-Purpose (FFP) cadastral mapping, particularly in resource-constrained environments where traditional, high-accuracy methods may be impractical. FFP approaches emphasize affordability, flexibility, and speed while ensuring that accuracy remains within an acceptable range for land tenure security.

Rocha et al. (2021) developed a quality assurance model for Fit-for-Purpose Land Administration in Colombia that aligns with ISO 19,100-series standards by incorporating positional accuracy, logical consistency, completeness,

and temporal accuracy into a structured spatial data quality evaluation framework. Their model uses RMSE-based classification of positional accuracy (ranging from 0.1 m to 3 m) to assess whether spatial data conform to specified quality levels in a national FFP context.

In Nigeria, Taiwo et al. (2025) conducted a comparative analysis of traditional cadastral survey methods (total station, theodolite traverse, and RTK GNSS) with high-resolution aerial imagery. Their findings indicate that aerial photogrammetry can achieve an accuracy of  $\pm 3$  m, which aligns with FFP standards and makes it a viable alternative to ground-based methods. However, they note that accuracy can vary based on image resolution and environmental conditions.

Another study by Taiwo et al. (2024) assessed the fitness of multi-resolution UAV orthomosaics (0.05 m to 1 m GSD) for cadastral mapping in Ekiti State, Nigeria, demonstrating that positional accuracy tends to improve with finer spatial resolution, with RMSE values ranging from approximately 0.05 m in formal settlements to over 2.5 m in informal rural contexts. This highlights that while fit-for-purpose cadastral approaches do not demand centimeter-level precision, the choice of geospatial data resolution has a significant impact on mapping reliability, particularly in areas lacking clear boundary markers.

Collectively, these studies indicate that although fit-for-purpose (FFP) cadastral mapping emphasizes affordability, scalability, and inclusivity, positional accuracy must remain appropriate to the intended land administration purpose and local context. Rather than adhering to a rigid universal threshold, FFP initiatives benefit from clearly defined quality assurance frameworks, suitable geospatial technologies, and sensitivity to settlement type, land use, and boundary visibility. Achieving this balance ensures spatial reliability while maintaining cost-effectiveness and operational efficiency.

Despite the growing body of literature on accuracy requirements for fit-for-purpose (FFP) cadastral mapping, there remains a significant gap in empirically validating the suitability and reliability of smartphone-based GNSS mobile applications for FFP cadastral mapping. While existing studies have explored smartphone positioning for general geospatial applications, limited research has systematically compared the accuracy of various GNSS mobile apps against FFP cadastral accuracy thresholds. Moreover, no comprehensive study has evaluated which specific applications are most suitable for FFP cadastral mapping, particularly in contexts where high-end survey-grade accuracy is not required. This research aims to bridge this gap by assessing the accuracy of multiple GNSS mobile applications and benchmarking their performance against established FFP accuracy standards, thereby contributing to the



body of knowledge on affordable, accessible, and scalable geospatial solutions for land tenure documentation, further advancing the practical implementation of FFP cadastral mapping in resource-constrained environments.

## Methods and datasets

This study utilized two categories of datasets. The first category comprised planimetric positional coordinates (X and Y) referenced to the WGS 1984 datum for six pre-marked survey stations within the Federal University of Technology, Minna (FUTMinna) campus. These coordinates were acquired using ten selected Android GNSS applications, each installed on Android-based smartphones of the same make and model. Each application independently recorded positional data at the survey stations to assess variations in accuracy and consistency. The second dataset consisted of reference coordinates for the same six survey stations, obtained using a high-precision Differential GNSS (DGNSS) system, a Hi-Target Differential GPS (DGPS) system consisting of a base station and rover unit, operating in static mode with a one-hour occupation time per station. The DGNSS data was post-processed using Hi-Target Geomatic Office (HGO) software to refine the positional coordinates and minimize errors.

The smartphones used in this study were equipped with built-in GNSS receivers capable of receiving and processing

signals from multiple satellite constellations, including GPS, GLONASS, and BeiDou. These devices supported the installation and concurrent operation of all ten GNSS applications.

For data processing and analysis, various software tools were employed. Hi-Target Geomatic Office (HGO) was used for post-processing DGNSS observations, improving coordinate accuracy through differential corrections. R-Suite and SPSS, statistical computing packages, were utilized for analyzing positional accuracy, computing errors, and evaluating the statistical significance of differences between smartphone GNSS and DGNSS coordinates.

## Study site

The study was conducted at the Federal University of Technology, Minna (FUT Minna), which is located in Niger State, Nigeria. FUT Minna is one of Nigeria's leading specialized Universities. The university spans two campuses: Gidan Kwano Campus and Bosso Campus, both of which are situated in different parts of Minna, the capital city of Niger State. The study was conducted at the Gidan Kwano Campus located approximately at Latitude ( $9.53^{\circ}$  N) and Longitude ( $6.47^{\circ}$  E). Fig. 1 presents a map highlighting the study area.

The Gidan Kwano Campus, located along the Minna–Bida Road, is the main campus and houses most academic faculties, administrative offices, and student accommodations. It

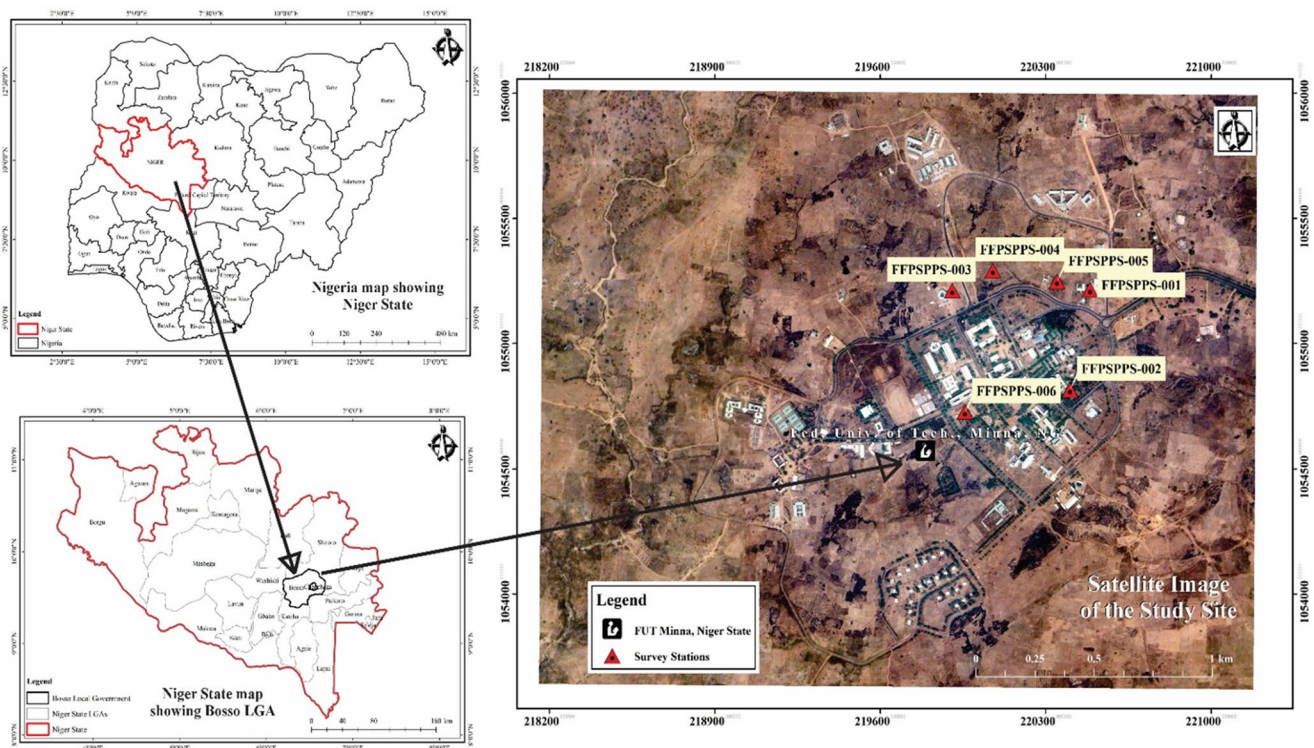


Fig. 1 Study area map

covers a vast expanse of land, making it an ideal location for spatial and geospatial studies where the study was carried out.

### Field data acquisition and processing

For smartphone positioning, ten different mobile GPS applications were installed on Android smartphones and simultaneously initialized to acquire positional data for six pre-marked survey stations. Data collection was conducted under favorable weather conditions to minimize atmospheric interference, particularly tropospheric delays.

Before data acquisition, essential parameters were configured. All GNSS observations were referenced to the World Geodetic System 1984 (WGS 84) datum, consistent with the International Terrestrial Reference Frame (ITRF) realizations adopted within the African Geodetic Reference Frame (AFREF) framework. The positional data were projected to the WGS 84/UTM Zone 32 N coordinate reference system (EPSG: 32632) to facilitate two-dimensional spatial analysis and linear deviation computation. Using an officially defined EPSG coordinate reference system ensures reproducibility, interoperability, and geodetic consistency of the results.

At each survey station, all ten mobile applications were activated and allowed to track an adequate number of healthy satellites. A stopwatch was used to monitor the occupation time, and each smartphone was carefully positioned over the survey beacon to ensure accurate alignment with the station. (Fig. 2).

To evaluate the performance of mobile applications in geographic positioning, four different occupation time intervals were initially set:

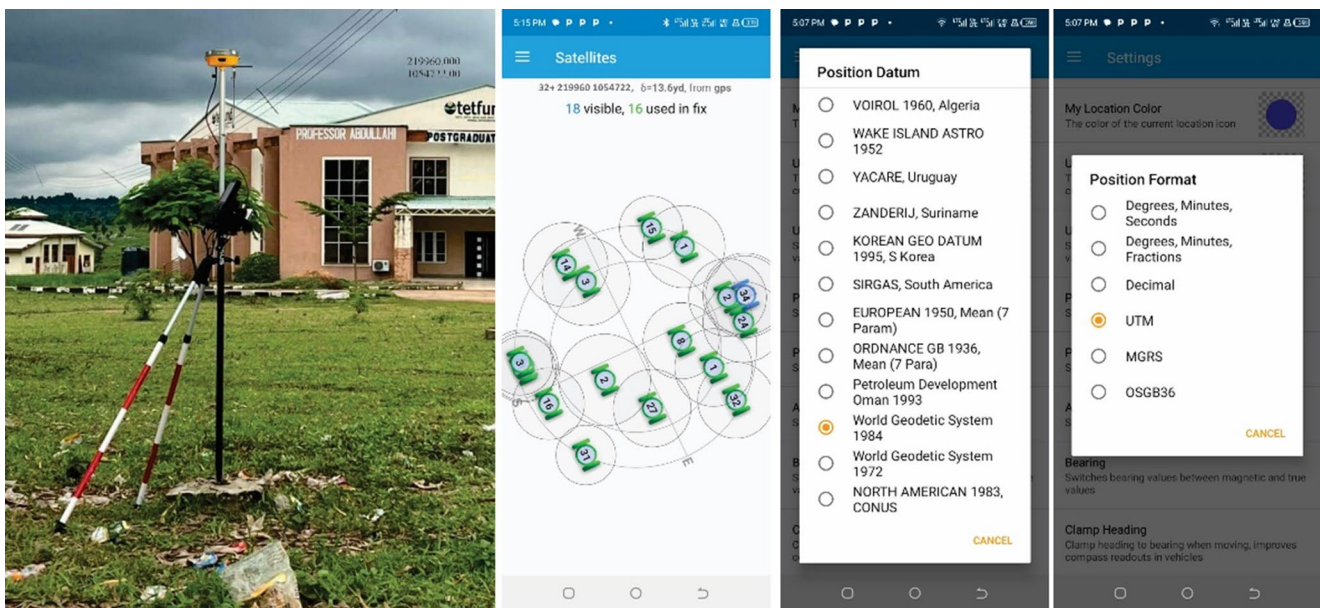
- 4 min (4').
- 15 min (15').
- 45 min (45').
- 60 min (1 h).

During the analysis, it was observed that increasing the occupation time beyond 4 min did not yield significant improvements in positional accuracy. As a result, observations from the 15-minute, 45-minute, and 60-minute intervals were filtered out, and only the 4-minute occupation time was retained for final evaluation.

Additionally, two applications, *Android GPSTest* and *Surveying*, were excluded from further analysis due to inconsistencies in their recorded positional data and their inability to meet the accuracy threshold required for Fit-For-Purpose (FFP) land administration. The final dataset, therefore, consists of positional coordinates obtained from the remaining mobile applications at the 4-minute occupation time.

Table 1 presents an excerpt of the refined field book used to record the smartphone-derived positioning data for this study, reflecting only the retained occupation time (4 min) and the selected GNSS applications.

For the static differential positioning technique, the Hi-Target DGNSS system was utilized. The base unit was set up at L-40, the reference station for the Minna Clarke 1880 datum, and configured with the necessary pre-observation



**Fig. 2** (a) DGNSS (rover) setup on FFPSPPS-006 (b) Position fixing with GPS Essential application (c) Setting of the Datum to WGS 1984 (d) Setting of the positioning format

**Table 1** Format and excerpt of the field book

| App:                            | Mobile     | GPS        | Polaris    | Handy                 | Topography                                 | GPS Location | GPS        | C7          |
|---------------------------------|------------|------------|------------|-----------------------|--------------------------------------------|--------------|------------|-------------|
|                                 | Topography | Essential  | Navigation | GPS                   |                                            |              | Data       | GPS         |
| Survey station ID.: FFPSPPS-001 |            |            |            |                       | Start Time: 09:15am                        |              |            |             |
| Date: 02/02/2025                |            |            |            |                       | Datum: WGS 84 (Zone 32)                    |              |            |             |
| E (m): 220486.931               | 220486.000 | 220486.800 | 220486.000 | 220485.2701055211.750 | 220485.00                                  |              | 220485.300 | 220485.266  |
| N (m): 1055214.132              | 1055214.00 | 1055214.20 | 1055214.00 |                       | 1055211.00                                 |              | 1055211.70 | 1055211.74  |
| Survey Station ID.: FFPSPPS-002 |            |            |            |                       | Start Time: 12:10pm                        |              |            |             |
| Date: 02/02/2025                |            |            |            |                       | Datum: WGS 84 (Zone 32)                    |              |            |             |
| E (m) 220404.695                | 220402.000 | 220403.600 | 220403.000 | 220403.6501054794.57  | 220403.00                                  |              | 220407.600 | 220407.198  |
| N (m) 1054812.660               | 1054789.00 | 1054820.20 | 1054795.00 |                       | 1054794.00                                 |              | 1054804.50 | 1054804.494 |
| Survey Station ID.: FFPSPPS-003 |            |            |            |                       | Start Time: 11:35amDatum: WGS 84 (Zone 32) |              |            |             |
| Date: 03/02/2025                |            |            |            |                       |                                            |              |            |             |
| E (m) 219907.407                | 219907.000 | 219907.300 | 219907.000 | 219907.4101055209.740 | 219907.000                                 |              | 219907.400 | 219907.407  |
| N (m) 1055209.738               | 1055209.00 | 1055209.80 | 1055210.00 |                       | 1055209.00                                 |              | 1055209.70 | 1055209.73  |
| Survey Station ID.: FFPSPPS-004 |            |            |            |                       | Start Time: 06:50pmDatum: WGS 84 (Zone 32) |              |            |             |
| Date: 03/02/2025                |            |            |            |                       |                                            |              |            |             |
| E (m) 220079.368                | 220080.000 | 220078.200 | 220071.000 | 220071.3701055297.060 | 220071.000                                 |              | 220071.400 | 220071.371  |
| N (m) 1055289.074               | 1055295.00 | 1055303.40 | 1055297.00 |                       | 1055297.00                                 |              | 1055297.10 | 1055297.056 |
| Survey Station ID.: FFPSPPS-005 |            |            |            |                       | Start Time: 12:30pmDatum: WGS 84 (Zone 32) |              |            |             |
| Date: 04/02/2025                |            |            |            |                       |                                            |              |            |             |
| E (m) 220349.671                | 220352.00  | 220350.400 | 220353.000 | 220352.7801055246.290 | 220352.000                                 |              | 220352.800 | 220352.779  |
| N (m) 1055247.055               | 1055246.00 | 1055247.40 | 1055246.00 |                       | 1055246.00                                 |              | 1055246.30 | 1055246.286 |
| Survey Station ID.: FFPSPPS-006 |            |            |            |                       | Start Time: 05:15pmDatum: WGS 84 (Zone 32) |              |            |             |
| Date: 05/02/2025                |            |            |            |                       |                                            |              |            |             |
| E (m) 219960.938                | 219960.000 | 219950.000 | 219950.000 | 219950.1101054719.89  | 219949.0001054719.00                       |              | 219950.100 | 219950.115  |
| N (m) 1054722.760               | 1054722.00 | 1054719.90 | 1054720.00 |                       |                                            |              | 1054719.90 | 1054719.88  |

settings (such as elevation mask of 10°) using the DGNSS receiver's data logger. Once properly initialized, the observation process commenced, with the start time duly recorded.

Simultaneously, the rover unit was deployed at each of the six survey stations. At each station, the receiver was allowed to acquire and process satellite signals for one hour to ensure optimal accuracy. The start and end times of each observation were recorded to maintain consistency in data collection. The longer observation duration was adopted to improve positional accuracy by allowing sufficient satellite visibility, improved satellite geometry, and mitigation of errors such as multipath and atmospheric effects through temporal averaging. Extended occupation times are commonly recommended for geodetic-grade GNSS observations when establishing reliable reference coordinates.

Upon completing data acquisition, the recorded positional data from both the base and rover units were downloaded and processed. Post-processing was carried out using Hi-Target Geomatics Office (HGO) software, which refined the raw observations and computed the final geodetic coordinates of the survey stations. Table 2 presents the DGNSS-derived coordinates of the six surveyed stations, along with the positional coordinates of the reference base station (L40) and the computed linear distances between the base station and each survey station.

Note that the DGPS observations were not conducted simultaneously with the smartphone GNSS measurements. Instead, the DGPS survey was first carried out to establish

**Table 2** The DGNSS resolved coordinates of the six surveyed stations

| Stations ID | Eastings (m) | Northings (m) | Distance (km) from base station |
|-------------|--------------|---------------|---------------------------------|
| L-40        | 227423.232   | 1066041.870   | -                               |
| FFPSPPS-001 | 220487.052   | 1055213.240   | 12.860                          |
| FFPSPPS-002 | 220402.844   | 1054814.633   | 11.768                          |
| FFPSPPS-003 | 219904.806   | 1055210.452   | 11.527                          |
| FFPSPPS-004 | 220076.084   | 1055290.847   | 11.400                          |
| FFPSPPS-005 | 220347.190   | 1055246.705   | 11.366                          |
| FFPSPPS-006 | 219958.086   | 1054726.146   | 11.975                          |

highly accurate reference coordinates for the selected survey stations. These coordinates served as the ground truth against which the positional accuracy of the smartphone GNSS mobile applications was evaluated.

The field data acquired from the survey stations using different mobile applications were analyzed by comparing them with the DGNSS-acquired coordinates. The linear distance deviation between the coordinates observed by each mobile application and the DGPS reference coordinates was computed. The deviation was determined using the Euclidean distance formula, which calculates the distance between two coordinated points based on changes in latitude (Northing) and departure (Easting), as expressed in Eq. (1):

$$l = \sqrt{(\Delta N)^2 + (\Delta E)^2} \quad (1)$$



where:  $l$ =Linear distance deviation of the mobile application from the DGNSS-observed position.

$\Delta N$ =Difference between the mobile application's observed Northing value and the DGNSS-observed Northing value.

$\Delta E$ =Difference between the mobile application's observed Easting value and the DGNSS-observed Easting value.

To further assess the accuracy of the eight GNSS mobile applications, descriptive and inferential statistical measures were employed, including mean and median linear deviations, standard deviation, 95% confidence intervals (CI), Root Mean Square Error (RMSE), and effect sizes (Cohen's  $d$ ) to quantify both the precision and practical significance of positional differences across applications. Equation (2) was used to compute 95% confidence intervals for the mean deviation of each application while Eqs. (3) and (4) was used in computing the Cohen's  $d$  which measures the effective size between two mobile GNSS application

$$CI_{95\%} = \bar{x} \pm t_{0.975, n-1} \cdot \frac{s}{\sqrt{n}} \quad (2)$$

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_{pooled}} \quad (3)$$

$$s_{pooled} = \sqrt{\frac{((n_1 - 1)s_1^2 + (n_2 - 1)s_2^2)}{n_1 + n_2 - 2}} \quad (4)$$

Where  $\bar{x}$  is the mean deviation of each application,  $s$  is the standard deviation of deviations,  $n$  is the number of stations,  $t_{0.975, n-1}$  is the t-value from student's t-distribution with  $n - 1$  degrees of freedom (for 95% confidence interval).

$\bar{x}_1$  and  $\bar{x}_2$  are the mean deviations of each application,  $n_1, n_2$  are sample sizes of the two groups,  $s_1, s_2$  are the standard deviations of the two groups.

## Results and discussions

### Observed trends in the mobile GPS accuracy

Several studies have established the potential of GNSS-based mobile applications in cadastral surveying and geospatial positioning. However, few have provided rigorous scientific validation of their applicability within the framework of Fit-For-Purpose Land Administration (FFPLA). This study aims to bridge that gap by not only evaluating the positional accuracy of selected mobile GNSS applications but also recommending the most suitable option, supported by statistical evidence. To assess the performance of the applications, the coordinate differences between each application's output and the reference DGNSS coordinates were computed, and their respective linear deviations analyzed. The deviations from each station were then summarized to obtain the mean linear deviation, standard deviation (with 95% CI) and median deviation for each application (see Table 3). Table 4 presents the standard deviation and Root Mean Square Error (RMSE) for each of the eight GNSS mobile applications, along with the corresponding effect sizes (Cohen's  $d$ ) computed relative to Mobile Topography, which demonstrated the best overall positional performance.

**Table 4** Mean, standard deviation, and median of the linear deviations

| Applications      | Std  | 95% CI      | RMSE  | Cohen's d<br>(vs. Mobile topography) |
|-------------------|------|-------------|-------|--------------------------------------|
| Mobile Topography | 1.20 | 1.57–4.09   | 3.04  | Reference                            |
| GPS Essential     | 9.06 | −2.06–16.96 | 11.13 | 6.46                                 |
| Polaris           | 4.63 | 1.04–10.77  | 7.26  | 3.38                                 |
| Handy GPS         | 6.67 | 0.86–14.86  | 9.94  | 4.79                                 |
| Topo App          | 6.58 | 1.20–15.02  | 10.09 | 4.73                                 |
| GPS Location      | 6.85 | 1.27–15.63  | 10.51 | 4.91                                 |
| GPS Data          | 3.73 | 2.72–10.54  | 7.45  | 2.77                                 |
| C7 GPS Dados      | 3.69 | 2.73–10.47  | 7.41  | 2.74                                 |

**Table 3** Mean, standard deviation, and median of the linear deviations

|                   | 001                                         | 002   | 003  | 004   | 005  | 006   | Mean                                 | Std  | 95% CI      | Med. |
|-------------------|---------------------------------------------|-------|------|-------|------|-------|--------------------------------------|------|-------------|------|
| Applications      | Linear deviation at station ID: FFPSPPS (m) |       |      |       |      |       | Descriptive statistical measures (m) |      |             |      |
| Mobile Topography | 0.90                                        | 2.71  | 2.70 | 3.73  | 2.51 | 4.43  | 2.83                                 | 1.20 | 1.57–4.09   | 2.71 |
| GPS Essential     | 1.30                                        | 25.65 | 2.63 | 5.71  | 4.86 | 4.57  | 7.45                                 | 9.06 | −2.06–16.96 | 4.72 |
| Polaris           | 0.99                                        | 5.62  | 2.58 | 12.73 | 3.28 | 10.22 | 5.90                                 | 4.63 | 1.04–10.77  | 4.45 |
| Handy GPS         | 1.30                                        | 19.63 | 2.24 | 7.98  | 5.85 | 10.16 | 7.86                                 | 6.67 | 0.86–14.86  | 6.92 |
| Topo App          | 2.32                                        | 20.08 | 2.70 | 7.80  | 5.61 | 10.14 | 8.11                                 | 6.58 | 1.20–15.02  | 6.71 |
| GPS Location      | 3.04                                        | 20.63 | 2.63 | 7.98  | 4.86 | 11.56 | 8.45                                 | 6.85 | 1.27–15.63  | 6.42 |
| GPS Data          | 2.33                                        | 11.19 | 2.70 | 7.81  | 5.62 | 10.14 | 6.63                                 | 3.73 | 2.72–10.54  | 6.72 |
| C7 GPS Dados      | 2.33                                        | 11.03 | 2.70 | 7.80  | 5.60 | 10.14 | 6.60                                 | 3.69 | 2.73–10.47  | 6.70 |



Table 4 demonstrates that Mobile Topography exhibited the lowest variability ( $SD=1.20$  m), narrowest confidence interval (1.57–4.09 m), and smallest RMSE (3.04 m), confirming its superior positional stability across the survey stations. In contrast, GPS Essential showed extremely high variability ( $SD=9.06$  m), a very wide confidence interval (−2.06–16.96 m), and the largest RMSE (11.13 m), indicating unstable and unreliable performance. The remaining applications (Polaris, Handy GPS, Topo App, GPS Location, GPS Data, and C7 GPS Dados) demonstrated moderate to high variability, with RMSE values ranging from 7.26 m to 10.51 m.

The computed effect sizes (Cohen's  $d$ ) further reveal substantial practical differences between Mobile Topography and all other applications, with effect sizes ranging from 2.74 to 6.46, indicating large to extremely large performance disparities.

Mobile Topography emerged as the most accurate and consistent application (see Fig. 3), exhibiting the lowest mean deviation standard deviation and RMSE among the evaluated applications. This indicates that the application produced the most reliable positional estimates under the observation conditions of this study. In contrast, applications such as Topo App and GPS Location displayed greater inaccuracies and variability. These results suggest potential limitations in their handling of environmental interferences and reflect possible weaknesses in their internal correction mechanisms.

While the mean deviation provides a quick indication of performance, a deeper insight emerges when examining the median values (Fig. 4). The median, being less sensitive to outliers, offers a more robust measure of central tendency, especially in datasets affected by inconsistent performance. Interestingly, when evaluated using median deviation, two additional applications achieved positional accuracies within a range that is operationally suitable for fit-for-purpose cadastral mapping, consistent with the accuracy levels (not more than 5 m) reported by Rocha et al. (2021) and Taiwo et al. (2025) for comparable survey contexts. This points to the presence of outliers in some applications' results and underscores the instability of certain apps under varying conditions.

It was observed that Mobile Topography's mean and median values are nearly identical, differing by only 12 cm. This consistency indicates minimal outlier impact, reinforcing the idea that the app benefits from some form of advanced internal correction. In contrast, other applications recorded discrepancies of up to 2 m between their mean and median values, reflecting less stable performance. Supporting this, the study by Kuboń and Plewako (2018) tested multiple smartphone applications for positioning accuracy and found that Mobile Topography produced the smallest

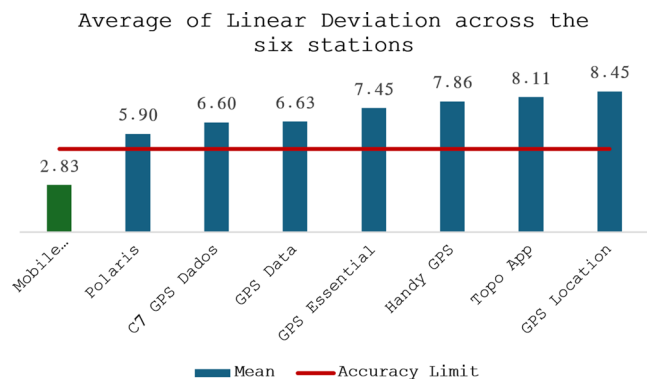


Fig. 3 Mean linear deviation of mobile applications across the six stations

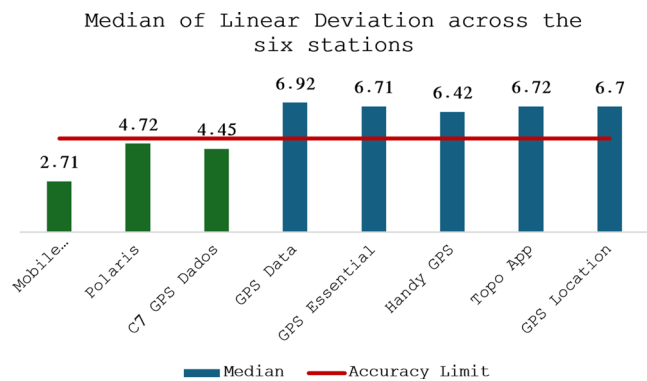


Fig. 4 Median of linear deviation across the six stations

average deviation (around  $\pm 2$  m), ranking it the most accurate in their evaluation.

According to the Mobile Topography GIS User Manual, the app's accuracy can be attributed to several built-in correction features:

- Weighted averaging on every axis, which improves location precision by averaging multiple GNSS readings and minimizing outlier influence.
- Map calibration, which aligns the device's GNSS data with known coordinates to correct systematic bias.
- Device calibration on a known point, allowing users to enhance measurement reliability by referencing fixed control points.

These features likely explain Mobile Topography's superior performance in this study and support its suitability for FFPLA-related cadastral activities.

The boxplot presented in Fig. 5 offers a visual representation of the performance variability of the mobile applications across the six survey stations. It reveals significant differences in both the accuracy and precision of the applications, highlighting areas where certain applications

**Fig. 5** Boxplot showing the variability and spread of linear deviation of the applications

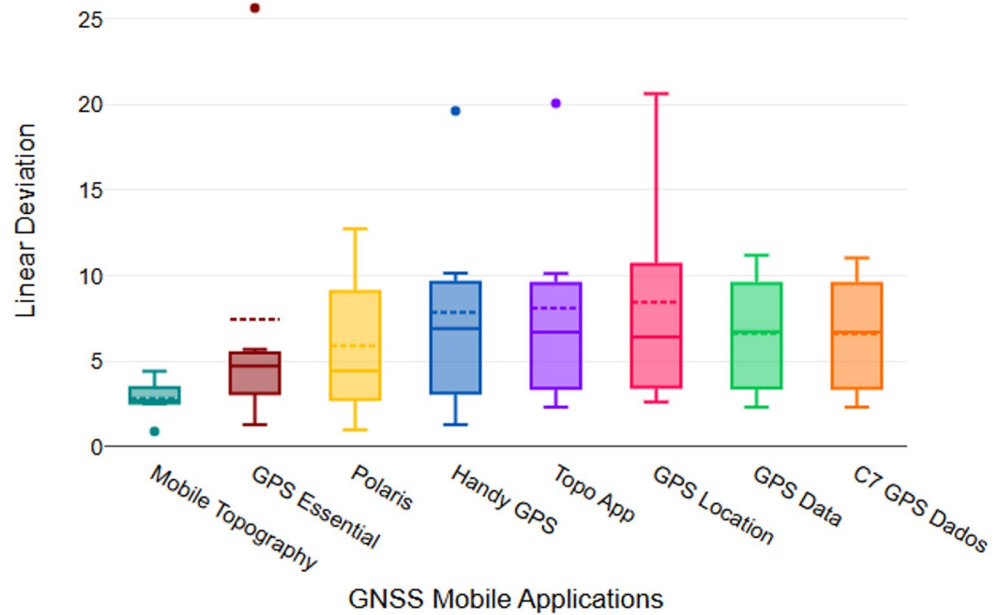


exhibit more consistency or substantial deviations in their positioning results.

A notable observation is the significant outlier in the performance of GPS Essential at Station 2, where a single outlier (25.65 m) severely skewed the application's overall performance. This outlier disproportionately affects the application's statistical measures, such as the mean and standard deviation. If this outlier were excluded from the dataset, the mean linear deviation of GPS Essential would reduce to 3.82 m, which would fall within the acceptable accuracy threshold of not more than 5 m for Fit-for-Purpose Land Administration (FFPLA) as recommended by previous studies (Rocha et al. 2021; Taiwo et al. 2025). This observation underscores the sensitivity of the application's accuracy to anomalous readings and highlights the potential impact of outliers on the overall evaluation of its performance.

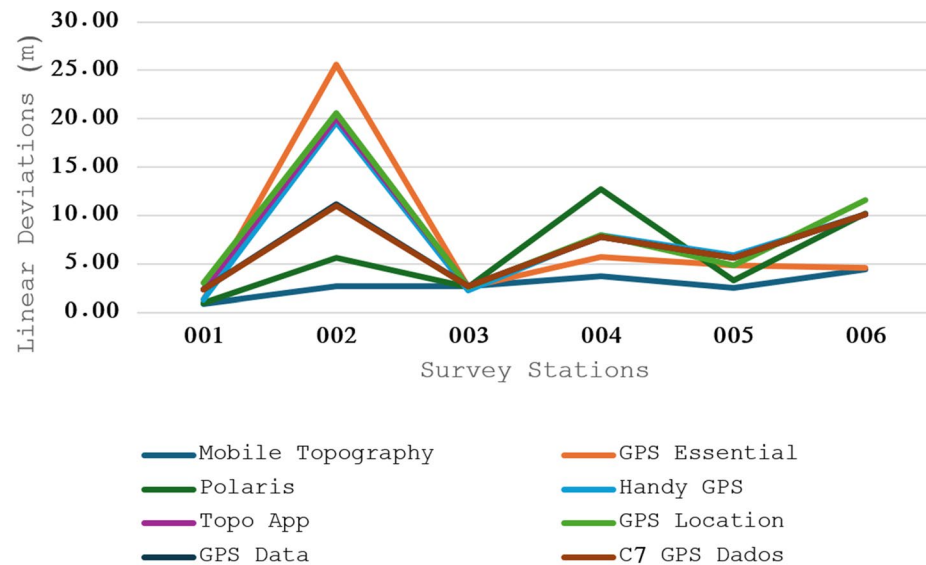
Interestingly, GPS Data and C7 GPS Dados exhibit a more uniform distribution of deviations across all six stations, suggesting that these applications are relatively stable and consistently reproduce similar results. This stability indicates that the applications may have robust algorithms for processing GNSS data, though their absolute accuracy still falls short of the required threshold for FFPLA. While the uniform distribution implies less variability in performance, the lack of sufficient accuracy suggests that these applications may require further refinement of their underlying error-correction algorithms to meet the precision requirements for FFP land administration.

The boxplot also indicates that five of the eight mobile applications tested exhibit poor performance due to outliers, which likely stem from the algorithms' inability to appropriately mitigate environmental and satellite signal anomalies. The presence of these outliers points to variability that

the applications failed to control, which is a critical concern in applications intended for land administration where satisfactory accuracy and consistency are paramount. Such performance variability emphasizes the need for further algorithmic advancements in these applications to handle diverse environmental conditions and correct errors more effectively.

For a more in-depth exploration of the stations where these outliers occurred, please see Fig. 6, which provides a detailed breakdown of station-specific deviations and highlights the stations with the most significant discrepancies in accuracy.

The presence of outliers in the performance of *GPS Essential*, *Handy GPS*, *Topo App*, *GPS Location*, *GPS Data*, and *C7 GPS Dados* occurred predominantly at Station 002. Stations 004 and 006 also exhibited similar deviations, although their influence on the overall results was comparatively minimal. These outliers significantly impacted the overall performance of the applications at these stations. The elevated linear deviations observed at Stations 002 and 006 are likely due to site-specific environmental conditions. Field inspection revealed the presence of nearby high-rise structures and surrounding vegetation at these locations, which may have introduced multipath effects and partial satellite signal obstruction. Such conditions are known to degrade GNSS positioning accuracy by causing signal reflections and attenuation. In contrast, Station 004 was located in a relatively open environment with minimal physical obstructions. The deviations observed at this station may, therefore, be associated with other factors such as temporary satellite geometry variations, signal quality fluctuations, or inherent limitations of smartphone GNSS receivers during the observation period.

**Fig. 6** Linear deviation of mobile applications

Additionally, the observed outliers may be related to limitations in smartphone-based GNSS signal reception. In some cases, the applications may have experienced difficulty receiving signals from a sufficient number of satellites due to reduced satellite visibility, weak signal strength, or the orientation of the device relative to the available constellations. This phenomenon is commonly reported in positioning applications that rely on the built-in GNSS receivers of smartphones, which generally provide lower signal tracking capability compared to geodetic-grade GNSS receivers used in conventional surveying equipment. Also, these outliers could stem from algorithmic weaknesses or insufficiencies in the mobile applications' data processing systems. Many mobile applications utilize algorithms designed to filter and correct for GNSS errors, but these algorithms can sometimes fail to properly handle complex scenarios where satellite signals are weak, intermittent, or subject to multipath effects. The failure to correct these issues in real-time can result in sporadic large deviations in the recorded coordinates.

It is important to note that these environmental conditions reflect realistic field scenarios encountered in cadastral data collection, where property boundaries may occur near buildings, vegetation, or other features that affect satellite signal reception. Consequently, the observed deviations highlight practical considerations that practitioners should account for when applying smartphone GNSS applications in fit-for-purpose cadastral surveying, rather than indicating deficiencies in the station selection process.

Interestingly, GPS Essential emerged as one of the more reliable applications tested in this study, showing promising results overall. However, the outlier at Station 2 substantially skewed its performance. Despite this single deviation,

GPS Essential still demonstrated solid potential for suitability in fit-for-purpose cadastral surveying, as its mean deviation (after excluding the outlier) would have comfortably met the accuracy threshold of 2-5m for FFPLA in rural areas which is typical of the study area. This suggests that, under typical conditions, the application could be a strong contender for land administration tasks, but further optimization and error correction algorithms would be needed to mitigate outliers and improve its consistency across varying environmental conditions. Polaris also produced promising results; however, it exhibited a similar issue to GPS Essential at the fourth station, indicating its tendency to yield variable coordinates depending on the prevailing conditions at the time of observation.

### Kruskal-wallis test

To assess whether differences in positional accuracy among the eight GNSS mobile applications are statistically meaningful or significant, a Kruskal–Wallis test was employed (See Table 5). This non-parametric test was chosen because the dataset exhibits non-normal distributions, contains outliers, and shows unequal variances across groups, violating key assumptions of parametric ANOVA. The test allows for a robust evaluation of whether any application performs significantly better or worse than the others, by comparing the median linear deviations across applications, providing insight into the relative reliability of each application for fit-for-purpose cadastral mapping.

As shown in Table 5 (a and b), the results indicated no statistically significant difference among the applications,  $\chi^2(7)=4.935$ ,  $p=0.668$ . Despite the lack of statistical significance, the mean rank table revealed that Mobile

**Table 5** Kruskal-Wallis test comparing the positional accuracy (linear deviation) of the eight smartphone GNSS mobile applications: (a) mean ranks and (b) test statistics.

| (a): Ranks           |                   |          |           |
|----------------------|-------------------|----------|-----------|
|                      | Applications      | <i>N</i> | Mean rank |
| Linear Deviation     | Mobile Topography | 6        | 13.75     |
|                      | GPS Essential     | 6        | 22.92     |
|                      | Polaris           | 6        | 23.75     |
|                      | Handy GPS         | 6        | 26.17     |
|                      | Topo App          | 6        | 26.83     |
|                      | GPS Location      | 6        | 29.58     |
|                      | GPS Data          | 6        | 26.92     |
|                      | C7 GPS Dados      | 6        | 26.08     |
|                      | Total             | 48       |           |
| (b): Test Statistics |                   |          |           |
| Statistics           | Linear Deviation  |          |           |
| Chi-Square           | 4.935             |          |           |
| Df                   | 7                 |          |           |
| Asymp. Sig. (p)      | 0.668             |          |           |

Topography consistently had the lowest mean rank (13.75), indicating the smallest positional deviations across the six survey stations. In contrast, GPS Location has the highest mean rank (29.58), reflecting larger deviations and greater variability. These findings, together with descriptive statistics (mean, standard deviation, RMSE) and effect size estimates (Cohen's *d*), demonstrate that Mobile Topography remains the most accurate and stable option for fit-for-purpose cadastral mapping, highlighting practical differences that may not be fully captured by inferential statistics alone.

The absence of statistically significant differences may be partly influenced by the presence of outliers and the limited number of observation stations, which can increase within-group variability and reduce the statistical sensitivity of the analysis. However, the stations served as controlled test points where multiple smartphone GNSS applications were evaluated against DGPS-derived reference coordinates. Descriptive statistics, including mean and median deviations, standard deviation, RMSE, and effect size (Cohen's *d*), provide robust evidence of relative performance.

## Conclusions

This study evaluated the positional accuracy and reliability of selected smartphone-based GNSS applications for Fit for Purpose Land Administration within a resource-constrained context. The findings demonstrate that Mobile Topography achieved the best overall performance among the tested applications, with a mean deviation of 2.83 m, thereby satisfying the FFPLA positional tolerance threshold of 2 to 5 m for rural cadastral applications. This result highlights the practical potential of low-cost mobile positioning technologies as viable alternatives in settings where access to

conventional high-precision geodetic equipment remains limited by financial or technical constraints.

Polaris and GPS Essential also exhibited moderate positional performance, suggesting that they may support selected land administration activities where very high cadastral precision is not required. However, the observed fluctuations in their accuracy indicate that smartphone-based GNSS performance remains sensitive to variations in satellite geometry, signal quality, environmental conditions, and the inherent processing limitations of mobile devices. Conversely, GPS Location consistently produced large positional deviations and irregularities, indicating limited suitability for boundary delineation and related cadastral operations.

While the study demonstrates the promise of smartphone-based GNSS applications for FFPLA implementation, certain limitations should be acknowledged. The analysis was restricted to a selected group of commercially available applications and specific smartphone devices under defined observational conditions. In addition, the proprietary nature of some commercial applications constrained deeper investigation into their internal processing algorithms, correction mechanisms, and reliability under varying operational environments. These limitations may affect the wider transferability of the findings across different devices, software ecosystems, and geographical contexts.

Notwithstanding these limitations, the study provides important empirical evidence on the operational feasibility of mobile GNSS technologies for cadastral data acquisition in developing and resource-constrained regions. The findings contribute to the growing discourse on affordable geospatial solutions for land administration and support ongoing global efforts toward more inclusive, accessible, and scalable FFPLA implementation strategies.

## Implications for future research

Future research should investigate advanced error-mitigation approaches, including satellite-constellation optimisation, weighted-averaging refinement, and integration with post-processing correction techniques, to improve positional reliability for cadastral applications. Further studies should also examine the influence of smartphone hardware variability, including GNSS chipsets, antenna configurations, and multi-frequency reception capabilities, on positional performance. Long-term experimental investigations across varying terrains, climatic conditions, and satellite visibility environments would also provide deeper insight into the operational stability of mobile GNSS applications. In addition, there is a need to assess more transparent and analytically flexible software environments, including open-source alternatives, that may enable deeper evaluation



of processing workflows and improve confidence in the deployment of FFPLA technologies.

**Author Contributions** O.G.A conceptualized the study and supervised it together with Y.D.O. G.A.A. and O.O.A performed the data analysis and wrote the main manuscript text which was revised by O.G.A and Y. D.O. All authors reviewed the manuscript.

**Funding** Open access funding provided by University of Pretoria. The authors did not receive support from any organisation for the submitted work, and no funding was received to assist with the preparation of this manuscript.

**Data availability** Data is provided within the manuscript or supplementary information files.

## Declarations

**Competing interests** The authors declare no competing interests.

**Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

## References

- Ajayi OG, Oruma E (2022a) On the applicability of integrated UAV photogrammetry and automatic feature extraction for cadastral mapping. *Adv Geodesy Geoinf (formerly Geodesy Cartography)* 71(1). <https://doi.org/10.24425/gac.2022.141172>
- AjayiOG, Oruma E (2022b) Integrated UAV-photogrammetry and automatic feature extraction for cadastral mapping. A featured article published in *Coordinates*, XVIII, issues 10 and 11. <https://mycoordinates.org/integrated-uav-photogrammetry-and-automatic-feature-extraction-for-cadastral-mapping>
- Ajayi OG, Bolarinwa OO, Odumosu JO, Kuta AA (2018) Fit-for-purpose cadastre: Development of a geospatial information software for digital land administration. Commonwealth Association of Surveying and Land Economics (CASLE) 2018 International Conference. Developing a sustainable professionalism in surveying & relevant education in the Commonwealth. Lusaka, Zambia. 29th November – 1st December
- Ajayi OG, Ajibade SA, Abdullahi AK (2021) Development of a geospatial information software for cadastral survey data processing and management. School of Environmental Technology International Conference, 3rd – 5th May, pp. F12 – F23
- Amir HP, Mohammad MAE (2024) Improving real-time smartphone positioning accuracy in urban environments. *J Geospatial Inform Technol* 12(3):45–62. <https://doi.org/10.1080/00396265.2016.1253522>
- Apostolopoulos K, Geli M, Petrelli P, Potsiou C, Ioannidis C (2018) A new model for cadastral surveying using crowdsourcing. *Surv Rev* 50(359):122–133. <https://doi.org/10.1080/00396265.2016.1253522>
- Asman N, Berahim N, Razali MN (2025) Optimizing land registration through a fit-for-purpose land administration framework. *International Journal of Research and Innovation in Social Science* 9(8):6893–6899. <https://doi.org/10.47772/IJRISS.2025.908000569>
- Bennett R, Unger E-M (2021) Fit-for-purpose land administration for all. *GIM Int*. <https://www.gim-international.com/content/blog/fit-for-purpose-land-administration-for-all>
- Chigbu UE, Bendzko T, Mabakeng MR, Kuusaana ED, Tutu DO (2021) Fit-for-purpose land administration from theory to practice: Three demonstrative case studies of local land administration initiatives in Africa. *Land* 10(5):476. <https://doi.org/10.3390/land10050476>
- Hernández Olcina J, Anquela Julián AB, Martín Furones ÁE (2024) Real-time cloud computing of GNSS measurements from smartphones and mobile devices for enhanced positioning and navigation. *GPS Solut* 28:167. <https://doi.org/10.1007/s10291-024-01705-8>
- Huang J, Guo Y, Li X, Zhang N, Jiang J, Wang G (2022) Evaluation of positioning accuracy of smartphones under different canopy openness. *Forests*. <https://doi.org/10.3390/f13101591>
- Kuboń N, Plewako M (2018) Examining the accuracy of positioning using selected smartphone applications. *Geomatics, Landmanagement and Landscape* 4:65–75. <https://doi.org/10.15576/GLL/2018.4.65>
- Mutale M, Ajayi OG (2026) Gender responsive spatial analysis of climate impacts on land access in Windhoek. *African Journal on Geospatial Sciences and Land Governance. Afr J Land Policy Geospatial Sci* 9(2):308–339. <https://doi.org/10.48346/IMIST.PRSM/ajlp-gs.v9i2.63096>
- Oluwadare C, Salami M (2024) Comparative Analysis of Smartphones and Survey-Grade GNSS Receivers for Parcel Boundary Determination. *J Appl Sci Technol Trends* 5(01):01–09. <https://doi.org/10.38094/jastt501179>
- Retscher G, Weigert T (2022) Assessment of dual-frequency multi-GNSS smartphones for surveying applications. *Appl Geomatics* 14:765–784. <https://doi.org/10.1007/s12518-022-00467-7>
- Rocha LA, Montoya J, Ortiz A (2021) Quality assurance for spatial data collected in fit-for-purpose land administration approaches in Colombia. *Land*. <https://doi.org/10.3390/land10050496>
- Strecha C, Rehak M, Cucci D (2024) The accuracy of RTK enabled mobile phones in RTK denied areas. *ISPRS Ann. Photogramm. Remote Sens. Spatial Inf. Sci.* X-4–2024:327–332. <https://doi.org/10.5194/isprs-annals-X-4-2024-327-2024>
- Taiwo IO, Ibitoye MO, Oladejo SO, Koeva M (2024) Fitness of multi-resolution remotely sensed data for cadastral mapping in Ekiti State, Nigeria. *Remote Sens*. <https://doi.org/10.3390/rs16193670>
- Taiwo IO, Ibitoye OM, Oladejo SO (2025) Comparison between aerial imagery and conventional cadastral mapping methods in Ekiti State, Nigeria; towards a fit-for-purpose approach. *Surv Rev* 57(402):244–257. <https://doi.org/10.1080/00396265.2024.2390237>

**Publisher's Note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.